



**KOLEJ MATRIKULASI SELANGOR
MATHS SURVIVOR
SEMESTER 1, SESSION 2020/2021**

**MATHEMATICS
SM015
2 HOURS**

NAME : _____
PRACTICUM : _____

INSTRUCTION:

1. This paper consists of **Part A (3 questions) & Part B (7 questions)**.
2. Answer **all** questions.
3. Your answers must be written with a black or blue pen only.
4. All steps must be shown clearly.
5. Please write your full name and class at the top of the answer sheets before handing in.
6. Please submit your answer sheets in PDF form.

PART A

(25 marks)

1. Evaluate the following limits, if exist :-

(a) (i) $\lim_{x \rightarrow 4} \frac{16 - x^2}{\sqrt{x} - 2}$

(3 marks)

(ii) $\lim_{x \rightarrow -\infty} \left(7 - \frac{\sqrt{5x}}{x} \right)$

(3 marks)

(b) If $\lim_{x \rightarrow 1} \frac{f(x) - 2}{3 - x} = 5$, find $\lim_{x \rightarrow 1} f(x)$.

(3 marks)

2. (a) Find the derivative of $f(x) = \frac{1}{\sqrt{2-x}}$ by using the first principle.

(4 marks)

(b) Find $\frac{dy}{dx}$ for each of the following:

(i) $y = \ln \sqrt{(4x-5)^3}$

(4 marks)

(ii) $y = \cot^3(7x^2 + 4x)$

(3 marks)

3. Find the relative maximum or minimum point for $f(x) = \frac{2}{x^2 + 5}$ by using first derivative test.

(5 marks)

PART B

(75 marks)

1. Given a complex number $z = \frac{(2-i)^2}{1+i}$.
- (i) Express z in the Cartesian form $a + bi$, where a and b are real numbers. (4 marks)
 - (ii) Find the modulus and argument of z . (4 marks)
 - (iii) Express z in polar form. (1 mark)
2. Find the interval of x for the following inequalities.
- (a) $\frac{8}{x+4} - 2 \leq 0$. (3 marks)
 - (b) $\left| \frac{3x+2}{x-1} \right| \geq 2$. (8 marks)
3. Given that $f(x) = x^2 - 2x - 3$, $x \leq 1$,
- (i) Find $f^{-1}(x)$. (3 marks)
 - (ii) Sketch the graphs of f and f^{-1} on the same axes. (3 marks)
 - (iii) State the domain and range of f^{-1} . (2 marks)
4. Given $f(x) = \ln(3x+2)$ and $g(x) = e^x + 2$.
- (i) Find $f^{-1}(x)$. (3 marks)
 - (ii) Find $g^{-1}(x)$. (3 marks)
 - (iii) Hence, find $(g \circ f)^{-1}(x)$. (4 marks)
5. The function f is defined as
- $$f(x) = \begin{cases} \frac{x^2 + 3x - 4}{|x-1|} & , \quad x < 1 \\ 5 & , \quad x = 1 \\ 2x - 7 & , \quad x > 1 \end{cases}$$
- (a) Find $\lim_{x \rightarrow 1} f(x)$. Hence, determine whether $f(x)$ is continuous at $x = 1$. (6 marks)

- (b) Determine the values of p and q such that the function

$$g(x) = \begin{cases} e^x + 2p & , \quad x < 0 \\ 3x^2 + 2x - 1 & , \quad 0 \leq x < 3 \\ -8q & , \quad x \geq 3 \end{cases}$$

is continuous on the interval $(-\infty, \infty)$.

(5 marks)

6. (a) Given that $\ln y = x^2 e^{3x} + 4y$, find $\frac{dy}{dx}$.

(6 marks)

- (b) A curve is defined by the parametric equations

$$x = 2t^3 \quad \text{and} \quad y = t^2 - 4t$$

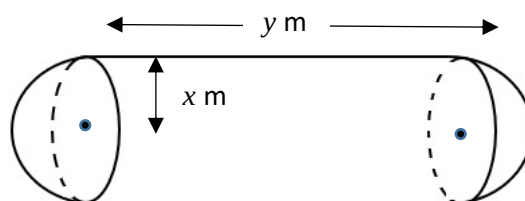
- (i) Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ in terms of t .

(6 marks)

- (ii) Find the point (x, y) on the curve where $\frac{dy}{dx} = 0$ and evaluate $\lim_{x \rightarrow \infty} \frac{d^2y}{dx^2}$.

(4 marks)

7. A solid is to be constructed in the shape of a cylinder of radius x m and length y m, with two hemispheres attached, one to each end of the cylinder as shown in the diagram below.



- (a) If the total surface area of the solid is $A \text{ m}^2$, express A in terms of x and y .

(2 marks)

- (b) If the volume of the whole solid is $\frac{\pi}{3} \text{ m}^3$, show that $y = \frac{1-4x^3}{3x^2}$.

(2 marks)

- (c) Hence, show that $A = \frac{2\pi}{3x} + \frac{4\pi}{3}x^2$ and find the radius of the cylinder such that the total surface area is minimum and hence find the minimum total surface area.

(6 marks)

END OF QUESTIONS PAPER